Part 1: Project Proposal and Dataset

Due November 12, 2026 · 100 points

Criterion	Excellent	Satisfactory	Needs Improvement	Unacceptable / Missing
Project Description and Goals (20 pts)	20: Clearly and thoroughly describes the application's purpose, intended functionality, user interaction, and use of the dataset. Game objectives, scoring, and win/lose conditions are explained if applicable.	16: Describes the project's goals and functionality adequately, but some details are missing.	10: Project description is vague, incomplete, or leaves significant questions about how the application will work.	0: Description is missing or does not establish a viable project.
Selection and Justification of Two Data Structures (25 pts)	25: Identifies two approved data structures, clearly justifies each selection, and accurately describes at least two meaningful operations for each.	20: Identifies two appropriate structures and their purposes, but explanations or operations lack some detail.	13: Structures are identified, but their purposes are unclear, justifications are weak, or meaningful operations are inadequately described.	0: Required structures are missing or selections do not satisfy assignment requirements.
Three Planned Application Features (15 pts)	15: Clearly describes three distinct, meaningful features, with at least two directly using the required data structures.	12: Describes three appropriate features, but some explanations lack detail or clarity.	8: Fewer than three meaningful features are adequately described, or their relationship to the data structures is unclear.	0: Features are missing or do not satisfy project requirements.
Dataset Selection (15 pts)	15: Provides an appropriate CSV dataset containing at least 2,000 records with at least three meaningful fields per record.	12: Provides an appropriate dataset meeting minimum requirements, but minor issues or inconsistencies exist.	8: Dataset is provided but has significant deficiencies in size, fields, format, or relevance.	0: Dataset is missing, unusable, or fundamentally inappropriate.
Detailed Project Task Chart (15 pts)	15: Provides a comprehensive, realistic task breakdown with responsibilities, due dates, and progress status for completed and remaining tasks.	12: Includes the required task chart information, but some tasks or details are insufficiently developed.	8: Task chart is substantially incomplete, lacks sufficient detail, or presents an unrealistic schedule.	0: Task chart is missing or provides little usable planning information.
Proposal Quality and Submission Requirements (10 pts)	10: Professional, organized proposal of approximately two pages; identifies partner arrangements and includes all required files in the correct ZIP structure.	8: Proposal is clear and generally organized with minor omissions or formatting issues.	5: Significant organizational, formatting, or submission deficiencies.	0: Proposal is missing or substantially fails submission requirements.
Total				100

Part 2: Nearly Complete Application

Due November 19, 2026 · 100 points

Criterion	Excellent	Satisfactory	Needs Improvement	Unacceptable / Missing
Overall Program Functionality and Progress (30 pts)	30: Program compiles, runs, and demonstrates substantial functionality. Application is nearly complete, with only minor remaining work.	24: Program compiles and runs with most major functionality implemented. Some additional development or debugging is necessary.	15: Program runs partially or has significant incomplete features. Substantial additional development is necessary.	0: Program is missing, cannot demonstrate meaningful functionality, or has made very little progress.
First Required Data Structure (15 pts)	15: Structure is appropriately implemented, meaningfully integrated, and demonstrates the required operations or substantial progress toward them.	12: Structure is implemented and used, but some operations or integration remain incomplete.	8: Structure is partially implemented or its practical use is limited or unclear.	0: Structure is missing or not meaningfully implemented.
Second Required Data Structure (15 pts)	15: Structure is appropriately implemented, meaningfully integrated, and demonstrates the required operations or substantial progress toward them.	12: Structure is implemented and used, but some operations or integration remain incomplete.	8: Structure is partially implemented or its practical use is limited or unclear.	0: Structure is missing or not meaningfully implemented.
CSV Dataset Integration (15 pts)	15: Program successfully loads and processes appropriate CSV data meeting the assignment requirements.	12: CSV loading and processing work with minor issues or incomplete integration.	8: CSV functionality is partially implemented or has significant limitations.	0: CSV integration is missing or nonfunctional.
README Documentation (15 pts)	15: README includes all required information, clear compilation/run instructions, data structure descriptions, known issues, and remaining tasks with responsibilities.	12: README includes most required information but lacks some details or clarity.	8: README is missing several important elements or provides insufficient instructions or progress documentation.	0: README is missing or largely unusable.
Code Quality and Submission Organization (10 pts)	10: Code is readable, logically organized, appropriately commented, and follows course programming practices. Required files and folders are correctly organized.	8: Code is generally readable and organized, with minor style, commenting, or submission issues.	5: Significant code organization, documentation, or submission problems.	0: Code is severely disorganized, required files are largely missing, or submission is unusable.
Total				100

Part 3: Completed Programming Project

Due December 1, 2026 · 100 points

Criterion	Excellent	Satisfactory	Needs Improvement	Unacceptable / Missing
Overall Program Functionality (15 pts)	15: Program compiles and operates correctly, accomplishes its intended purpose, handles invalid input appropriately, and permits normal termination.	12: Program functions correctly overall, with only minor errors or limitations.	8: Program has significant functional problems, unreliable behavior, or inadequate input handling.	0: Program is missing, cannot run, or fails to provide meaningful functionality.
Three Meaningful Application Features (15 pts)	15: All three features are implemented and function correctly. Features are distinct and meaningful, with at least two directly using the required structures.	12: Three appropriate features are implemented, but minor functionality or integration issues exist.	8: One or more features are incomplete, unreliable, or insufficiently meaningful.	0: Required features are largely missing or nonfunctional.
First Required Data Structure (10 pts)	10: Structure is appropriately selected and meaningfully integrated, with at least two correctly implemented operations that support application functionality.	8: Structure is appropriately used, but minor operational or integration issues exist.	5: Structure is implemented but has significant limitations, insufficient operations, or weak integration.	0: Structure is missing, invalid, or not meaningfully used.
Second Required Data Structure (10 pts)	10: Structure is appropriately selected and meaningfully integrated, with at least two correctly implemented operations that support application functionality.	8: Structure is appropriately used, but minor operational or integration issues exist.	5: Structure is implemented but has significant limitations, insufficient operations, or weak integration.	0: Structure is missing, invalid, or not meaningfully used.
CSV Dataset Integration (10 pts)	10: Dataset satisfies all requirements, loads successfully, and contributes meaningfully to application functionality.	8: Dataset meets requirements and is used appropriately, with minor processing/integration issues.	5: Dataset has significant deficiencies or is only minimally incorporated into the application.	0: Required dataset is missing, invalid, or not meaningfully used.
Code Quality and Organization (10 pts)	10: Code is well organized, readable, appropriately documented, and follows sound programming practices.	8: Code is generally well structured and readable, with minor style or documentation deficiencies.	5: Code has significant problems involving organization, readability, commenting, or prog. practices.	0: Code is disorganized, lacks meaningful documentation.
Final Project Report (15 pts)	15: Complete, well-written 2–4-page report covering all required sections, accurate data structure explanations and diagrams, lessons learned, AI documentation, and five meaningful test cases.	12: Covers all major requirements but has minor omissions, weak explanations, or documentation deficiencies.	8: Multiple required sections are incomplete, or data structure explanations, diagrams, testing, or AI documentation are substantially inadequate.	0: Report is missing or substantially fails to satisfy requirements.
Individual Video Demonstration (10 pts)	10: Clear, independently recorded 2–5-minute demonstration showing all three features, CSV usage, data structure operations, and accurate explanations of structure selection.	8: Demonstration covers most requirements but contains minor omissions or weak explanations.	5: Demonstration omits significant functionality, provides inadequate explanations, or does not adequately demonstrate the student's understanding.	0: Demonstration is missing, unusable, or does not meaningfully demonstrate the project.
README and Submission Requirements (5 pts)	5: Updated README and all required source files, data, documentation, and proof screenshot are correctly organized and submitted.	4: Most requirements satisfied, with minor omissions or organization problems.	2: Multiple submission requirements are incomplete or incorrectly organized.	0: Major required deliverables are missing or submission is largely unusable.
Total				100